k_2 is the rate constant for the dissociation of ES complex into E & S and

k_3 is the rate constant for the dissociation of ES complex into P.

K_m is a measure of the strength of ES complex. A high value of K_m indicates weak binding and a low value of K_m indicates strong binding between enzyme and substrate. The K_m values of a few enzymes are given below.

ENZYME	SUBSTRATE	K_m(mM)
Aspartate aminotransferase	Aspartate α-Ketoglutarate Oxaloacetate Glutamate	0.9 0.1 0.04 4
Threonine deaminase	Threonine	5
Arginyl-tRNA synthetase	Arginine $tRNA^{Arg}$ ATP	0.003 0.0004 0.3
Pyruvate carboxylase	HCO_3^- Pyruvate ATP	1.0 0.4 0.06
Penicillinase	Benzylpenicillin	0.05
Lysozyme	Hexa-*N*-acetylglucosamine	0.006

Note: For a given enzyme K_m is a constant for a specific substrate.

9.1.3. Lineweaver - Burk Equation:

The Michaelis - Menten equation is

$$v = \frac{(V_{max}[S])}{(K_m + [S])} \quad \ldots(1)$$

Taking reciprocals on both sides of equation (1)

$$\frac{1}{v} = \frac{K_m}{V_{max}} \frac{1}{([S])} + \frac{1}{V_{max}} \quad \ldots(2)$$

This is the equation for a straight line., $y = mx + c$

Equation (2) is known as Lineweaver - Burk equation.

A plot of $\frac{1}{v}$ versus $\frac{1}{([S])}$ is called a double - reciprocal plot (or) Lineweaver - Burk plot (LB plot) which yields a straight line.

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